Mentor Prep — CNN-GRU-Attention vs ImgFi

Everything you need to defend Action_NoAction_CNN_GRU.ipynb, with every number measured from our own data. Study Parts 1–5; Part 8 is the rapid-fire bank.

Golden rule for the whole meeting: every claim you make should be backed by a cell in the notebook you can scroll to. If you cannot point at it, do not claim it.

Part 0 — The 60-second opening

> "We captured 8 CSI takes on a USRP N210 — two people, `action` vs `noaction`, 322 windows of 2 seconds. We built two pipelines. The first follows ImgFi: turn each of the 52 subcarriers into a recurrence-plot image and classify all 52 with a ResNet-18, then majority vote. The second is ours: feed the whole 52×200 amplitude matrix to a CNN-GRU with attention in a single pass.
>
> Ours gets 98.4% cross-subject with 0.57 million parameters instead of 11.2 million, and one forward pass instead of 52. But the honest headline is this: **a single scalar feature with one threshold also gets 98.4%.** So this binary task does not prove our architecture is better — it proves the capture-to-evaluation chain works. I can show you exactly why we still believe the architecture is the right choice for the 7-class problem, and I measured the three things that justify it."

That framing does two things: it gets ahead of the "your accuracy is meaningless" objection, and it shows you know the difference between *validating a pipeline* and *validating a model*.

Part 1 — CSI fundamentals (she will start here)

What is CSI, physically?

When a WiFi transmitter sends an OFDM frame, the receiver estimates the channel from the known preamble. For each subcarrier *k* it recovers a complex number:

$$H_k = Re_k + j\, Im_k$$

$|H_k|$ is how much that frequency was attenuated; $\angle H_k$ is how much it was delayed. A person moving between transmitter and receiver changes the multipath environment, so H_k changes over time. **That time-variation is the entire sensing signal.**

Why exactly 52 subcarriers?

802.11a/g at 20 MHz uses a 64-point FFT → 64 subcarrier slots. Of these, 12 are unused (DC null + guard bands at the edges), leaving **52 usable subcarriers** (48 data + 4 pilot). This is dictated by the standard — **we did not choose 52**, it is what the PHY gives us.

- Intel 5300 NIC datasets (UT-HAR, Widar 3.0, SignFi) report only **30** subcarriers because the NIC firmware reports a subsampled subset.
- **We report all 52** — finer frequency resolution than the Intel-based datasets.
- ARIL (also USRP-based) also has 52. Our front-end matches theirs.

> **If asked "why not 64?"** → 12 are null by standard (DC + guards); there is no channel estimate to report for them.

Why amplitude only? Why throw away phase?

This is the most likely first hard question. Full answer:

Our USRP is *not* synchronised to the MacBook. Three independent offsets corrupt the phase of every packet:

Offset	Cause	Effect on phase
CFO (carrier frequency offset)	TX and RX local oscillators differ	rotation growing over the packet
SFO (sampling frequency offset)	ADC clocks differ	subcarrier-dependent linear ramp
PBD / timing offset	packet detection lands a few samples off	linear ramp across subcarriers

These change **randomly packet to packet**, so the raw phase is effectively noise. Amplitude $|H| = \sqrt{Re^2 + Im^2}$ is invariant to all three (they are pure phase rotations).

Two strong supporting arguments:

1. **ARIL discards phase too** — and they had *two* USRPs with an OctoClock GPS-disciplined clock and controlled both ends. If a fully synchronised setup still drops phase, our passive-sniffing setup certainly should.
2. **Widar 3.0 uses phase, but only phase *differences between antennas* on the same NIC.** The offsets are common to all antennas on one card, so subtracting cancels them. **That trick needs ≥2 receive antennas. We have one (SISO), so it is impossible for us in principle** — not merely unimplemented.

> **If she says "phase carries more information, papers use it"** → agree, then explain the antenna-difference requirement above. Then say: *"adding a second RX antenna is on our upgrade list precisely because it unlocks phase-difference features like Widar's DFS/BVP."* That turns a weakness into a roadmap item.

What does one sample look like?

- One row of the CSV = one decoded 802.11 frame = `Packet, Timestamp_us, SC0_Re, SC0_Im, … SC51_Re, SC51_Im` (106 columns).
- We capture at **51.5–77.6 Hz** (mean **66.4 Hz**) — this is *packets per second*, i.e. our sampling rate.
- One training sample = **2 seconds = 52 subcarriers × 200 time steps**.

Part 2 — Data pipeline, step by step (§1–§5 of the notebook)

Step	What happens	Why
1. Capture	USRP passively sniffs MacBook→hotspot frames; `extract_csi` filters by MAC and writes CSV	Passive = no control over rate; realistic deployment scenario
2. Amplitude	$\|H\| = hypot(Re, Im)$	Phase unusable (Part 1)
3. Windowing	2 s spans, **resampled onto a uniform grid using real timestamps** → exactly 52×200	Packets arrive irregularly (median gap 12 ms, worst 464 ms). Interpolating onto a time grid makes every window exactly 2 s of wall clock
4. Fill check	Skip window if <50% of expected packets	Prevents mostly-interpolated windows entering the dataset
5. Detrend	Subtract each subcarrier's temporal mean	Removes the large static component (room geometry, gain, distance) which carries no motion info; also gives robustness to slow gain drift
6. Scale	Divide by per-subcarrier std **from the training split only**	Subcarriers have different baseline variance (frequency-selective fading). Training-only avoids leakage

Numbers to have memorised

- **322 windows** total: 168 `action`, 154 `noaction` (52.2% action — balanced).
- By subject: amrith **182**, rishiikesh **140**.
- By take: walk 87, mixed 81, stand 81, sit 73.
- Each capture has **exactly 1 truncated final row** (Ctrl+C landed mid-write) — we drop it explicitly. 0.015% of data. Cosmetic.

> **Known flaw, disclose it yourself before she finds it:** we segmented with the pipeline's default `--rate 100` on 66 Hz data. A 2 s window then needs ≥100 packets to pass the fill check, so **48 of 370 possible windows (13%) were discarded**. Re-running with `--rate 50` recovers them with no recapture. Volunteering this makes everything else you say more credible.

Part 3 — Architecture, layer by layer (§7)

Exact shape trace (verify live in the notebook)

```
input                       (B, 52, 200)    B=batch, 52 subcarriers, 200 time steps
  Conv1d(52→64, k=5, pad=2)  (B, 64, 200)
  BatchNorm1d(64) + ReLU     (B, 64, 200)
  MaxPool1d(2)               (B, 64, 100)
  Conv1d(64→128, k=5, pad=2) (B, 128, 100)
  BatchNorm1d(128) + ReLU    (B, 128, 100)
  MaxPool1d(2)               (B, 128, 50)
transpose                    (B, 50, 128)    → (batch, time, features) for the GRU
  GRU(128→128, 2 layers, bidirectional)
                             (B, 50, 256)    256 = 128 hidden × 2 directions
  Linear(256→1) + softmax    (B, 50, 1)      attention weights, sum to 1 over time
  weighted sum over time     (B, 256)        context vector
  Linear(256→64)+ReLU+Dropout(0.5)+Linear(64→2)
                             (B, 2)          logits
```

Parameter budget (printed in the notebook)

Component	Parameters	Share
CNN feature extractor	58,176	10.2%
BiGRU	**494,592**	**86.8%**
Attention	257	0.05%
Classifier head	16,578	2.9%
Total	**569,603**	100%

ResNet-18 (the ImgFi notebook backbone) = ~11.18 M → **19.6× larger than ours**.

> **If she asks "where do the parameters actually go?"** → 87% is the GRU. Say you know that, and that it is the obvious first target if the model needs to shrink (1 layer instead of 2, or hidden 64 instead of 128).

THE key question: "Why is 52 the channel dimension?"

This is the question she is most likely to press on, and it is your best answer.

`nn.Conv1d(in_channels=52, out_channels=64, kernel_size=5)`:

- The **52 subcarriers are input channels**, like RGB is 3 channels in an image CNN.
- The kernel slides along **time only**.
- At each time step, every filter sees **all 52 subcarriers simultaneously**.
- So a filter can learn a pattern like *"subcarriers 10–20 rise while 35–45 fall"* — a **cross-subcarrier spatial signature** of the multipath geometry.

ImgFi structurally cannot do this. It makes one image per subcarrier, so each image sees exactly one subcarrier, and majority voting at the end combines *decisions*, not *features*. The joint pattern is destroyed by construction.

And we proved the joint pattern is real (§3):

Measurement	`action` takes	`noaction` takes
Mean |correlation| between subcarriers	0.38–0.45	0.18–0.40
Effective rank (of 52)	**3.8**	11.9

During motion the 52 subcarriers collapse to **≈4 independent dimensions**, because one moving body imposes *one* dominant channel perturbation that all subcarriers see together.

Two consequences — say both:

1. ImgFi's "×52 free data expansion" is not ×52 of *information*. The 52 images per window are ~4 distinct views plus 48 near-duplicates. The effective dataset is closer to 322 × 4 than 322 × 52.
2. The correlation *structure itself* is a feature — and it is exactly what our Conv1d reads and ImgFi discards.

> **Effective rank, if she asks the definition:** $exp(H(\lambda))$ where H is the Shannon entropy of the normalised eigenvalue spectrum of the covariance matrix. It is a continuous version of "how many dimensions actually matter". 52 equal eigenvalues → 52; one dominant eigenvalue → ≈1.

Layer-by-layer rationale

`Conv1d(52→64, kernel_size=5, padding=2)`

- k=5 at the 100 Hz window grid = **50 ms**, the timescale of a limb movement.
- `padding=2` keeps length at 200 (`same` padding) so nothing is lost at the edges.
- 64 filters = 64 different learned cross-subcarrier patterns.

`BatchNorm1d` — subcarrier variances differ by orders of magnitude (frequency-selective fading). Normalising per channel keeps gradients well-scaled and speeds convergence.

`ReLU` — nonlinearity; cheap, no vanishing gradient.

`MaxPool1d(2)` — halves time (200→100→50), keeps the strongest response in each pair, gives small shift-invariance. After both pools, **each GRU time step = 40 ms of real time**, and the CNN's receptive field is ≈13 original samples ≈ **130 ms**.

`Conv1d(64→128)` — second stage composes simple primitives into longer patterns (a classic CNN hierarchy).

`GRU(128→128, num_layers=2, bidirectional=True)`

- *Why any RNN?* The CNN only sees ±130 ms. An activity is an **ordered sequence** — squat down, pause, come up. The GRU carries state across the whole 2 s and can represent order.
- *Why GRU not LSTM?* 3 gates instead of 4 → ~25% fewer parameters. With only ~130 training windows per fold, the smaller model is the right call. LSTM's extra cell state buys little at a 50-step sequence.
- *Why bidirectional?* We classify a **complete, already-recorded** window offline, so the end of the window is legitimately available when interpreting the beginning. **If she asks about real-time deployment:** correct — bidirectional adds latency, and for streaming we would switch to unidirectional or run it on a sliding buffer. Acknowledge it.

- *Why 2 layers?* One layer underfits the compositional structure; three overfits 130 samples.

Attention: `Linear(256→1)` + softmax over time

- *The problem it solves:* motion does not fill the window. A squat may occupy 0.6 s of 2 s. **Mean-pooling** would dilute the informative 0.6 s with 1.4 s of idle. **Last hidden state** would discard it entirely if it happened early.
- *The mechanism:* score each of the 50 time steps, softmax to weights summing to 1, take the weighted average. The model **learns which moments matter**.
- *Cost:* 257 parameters. Essentially free.

And we verified it does something (§13):

- Attention peak / uniform = **2.22×** (1.0 would mean flat = identical to mean-pooling).
- Correlation between attention weight and instantaneous motion energy = **+0.346** across all 182 test windows.

> That is the answer to *"you added attention, prove it isn't decoration."* We measured it.

`Dropout(0.5)` — 322 windows is tiny; 0.5 is aggressive regularisation, deliberately.

`Linear(256→64→2)` — small head, no reason for more capacity.

Part 4 — Training and evaluation protocol (§9, §10)

Setting	Value	Why
Loss	CrossEntropyLoss	standard for classification
Optimiser	Adam, lr 1e-3	robust default; no time to tune on 322 samples
LR schedule	ExponentialLR γ=0.85	smooth decay; loss plateaus by ~epoch 10
Epochs	30	training accuracy saturates ~epoch 5; rest confirms stability
Batch	32	small dataset → small batch → more updates per epoch
Seed	fixed (0)	results are reproducible; the original run had no seed

The protocol question — Leave-One-Subject-Out

Say this clearly, it is a strength:

> "We never use a random split. All windows from one take are highly correlated — same person, same clothes, same position, seconds apart. A random split puts near-duplicates in both train and test and gives a meaningless 99%. We train on one person and test entirely on the other, in both directions, and we report the **worse** of the two."

Model selection uses a **validation set carved from the training subject's windows** (15%, split by window). **The test subject is never used to choose anything.**

> **We are stricter than the published datasets we compare against:** ARIL tests on "every 5th sample" from the *same single volunteer*; UT-HAR splits randomly. Only Widar 3.0 has a comparably clean protocol.

Part 5 — ImgFi: the paper, our notebook, and our model

She will ask *"the paper did X, why did you do Y?"* — so separate three different things.

5.1 What the ImgFi paper actually says

- Convert CSI time series to images; compares 5 methods (RP/RT, GASF, GADF, MTF, STFT).
- **RT (recurrence transform), $RT[i,j] = |x_i - x_j|$, scored best (~99.8%)** and is cheapest.
- **Core claim: a lightweight 3-conv-layer CNN matches ResNet-50/VGG-19 with ~90× fewer parameters.** The paper is fundamentally an *efficiency* argument.

5.2 What our ImgFi notebook actually did — and where it diverges

⚠️ **Know these before she does. They are all defensible if you raise them first.**

Aspect	ImgFi notebook	Issue
Backbone	**ImageNet-pretrained ResNet-18 (11.2 M)**	The paper's whole point is a *lightweight 3-conv CNN*. Using ResNet-18 **contradicts the paper's thesis** — so it is not a reproduction of ImgFi, it is a recurrence-plot pipeline with a heavy off-the-shelf backbone
Windowing	slices **100 packets** assuming 50 Hz	Real rate is 51.5–77.6 Hz, so a "2 s" window is really **1.29–1.94 s** and varies per file. Windows are not physically comparable
Duration reporting	`rows / 50`	Reports amrith `mixed` as **158.6 s** when it is really **107.9 s**
Normalisation	`StandardScaler` fit on **all 8 files**	Test-subject statistics leak into the scaler
Model selection	evaluates on the **test set** every epoch, keeps the best	Test set influenced selection → the 97.8% "best" is optimistically biased
LOSO folds	only **1** (test = amrith)	The rishiikesh direction was never run
Normalisation constants	ImageNet mean/std on a replicated grayscale RP	Arbitrary for recurrence plots (harmless, but she may ask)

How to present this: *"Our ImgFi notebook is a working recurrence-plot baseline, but it is not a faithful reproduction of the paper and it has three protocol issues we identified. That is precisely why we do not claim our model beats it on accuracy — the comparison isn't like-for-like yet."* That is a much stronger position than defending a flawed comparison.

5.3 Head-to-head

	ImgFi notebook	Ours
Representation	52 RP images, 224×224	raw 52×200 matrix
Subcarriers	**independent**, majority vote	**joint**, CNN input channels
Windowing	100 packets (assumes 50 Hz)	2 s wall-clock, timestamp-resampled
Normalisation	all files (leak)	train split only
Backbone	ResNet-18, 11.2 M	CNN-GRU-Attn, **0.57 M**
Forward passes/window	**52**	**1**
Temporal order	implicit (image diagonal)	explicit (BiGRU + attention)
Model selection	on test set	on validation split
LOSO folds	1	2
Accuracy	95.9% final / 97.8% best (amrith)	**98.4%** (amrith), 98.6% (rishiikesh)
Inference	52 × 224×224×3 passes	**0.55 ms**, one pass

Crucial caveat to state out loud: the two pipelines segment differently — ImgFi produces **497** windows of ~1.5 s average, we produce **322** windows of exactly 2 s. **Different sample sets, so the accuracies are not directly comparable.** What *is* comparable and solid: 19.6× fewer parameters, 52× fewer forward passes, correct time-based windowing, no leakage, honest model selection.

5.4 "So did you follow the paper or not?"

> "We follow the paper's *goal* — a lightweight model — but not its representation. The paper reached efficiency by keeping the image representation and shrinking the CNN. We reached it by dropping the image representation entirely: no 224×224 upsampling of a 200-point series, no 52× duplication of work. 0.57 M parameters is in the spirit of the paper's ~90× reduction claim. And we have a measurement the paper does not: on *our* hardware the 52 subcarriers carry only ~4 independent dimensions, which is what removes the benefit of treating them as 52 separate images."

Part 6 — Results, and the honest interpretation (§10–§15)

Headline

Fold	Test windows	Accuracy	Confusion (rows=true)
test = amrith	182	**98.352%**	action 91/1 · noaction 2/88
test = rishiikesh	140	**98.571%**	action 76/0 · noaction 2/62
Headline (worse)		**98.35%**	

Precision/recall for `action`: 0.978/0.989 (amrith), 0.974/1.000 (rishiikesh).

Per-activity breakdown — and a result that surprised us

Take	n	Accuracy
walk	87	1.000
sit	73	1.000
mixed	81	0.988
stand	81	**0.951**

We expected `mixed` to be hardest (it has the small drink/bend motions). It wasn't — **`stand` was.** All `noaction` errors come from *standing still*, none from *sitting still*.

Physical explanation: a standing person is never truly static — postural sway, weight shifts and breathing perturb the channel. A seated person is mechanically braced. So `stand` windows sit closest to the decision boundary.

> Reporting a disproved hypothesis *with an explanation* is exactly the kind of thing that impresses a mentor. Do not hide it.

The uncomfortable result — lead with it, don't get caught by it

Baseline: one scalar (mean over subcarriers of temporal std = "motion energy"), one threshold, chosen on the training subject only.

Fold	1-feature baseline	CNN-GRU-Attention	Gap
test = amrith	98.35%	98.35%	**0.00**

Fold	1-feature baseline	CNN-GRU-Attention	Gap
test = rishiikesh	**99.29%**	98.57%	**−0.72**

`action` windows carry **4.20×** the motion energy of `noaction` (0.01452 vs 0.00346).

The deep model does not beat a one-line threshold. On one fold it is slightly worse.

How to say it:

> "This does not mean the architecture is wrong — it means **this binary task is too easy to distinguish a good model from a trivial one**. Separating 'moving' from 'not moving' is a single-scalar problem. Our 98.4% validates the chain — capture, segmentation, subject- independent evaluation — not the architecture.
>
> The architecture's justification is *structural*, not this accuracy number: cross-subcarrier information exists and we exploit it (§3), our sampling rate covers the informative band (§4), attention is measurably non-uniform (§13), and we are 19.6× smaller with 52× fewer passes (§16). The real test is 7 classes, where a single energy scalar **cannot** separate `squat` from `bend` from `drink` — they have similar energy but different *temporal structure*, which is exactly what the GRU and attention encode."

Sampling rate — the strongest single piece of analysis (§4)

She may say *"UT-HAR and Widar sample at 1000 Hz, you sample at 66 Hz — your data is inadequate."* You have a measured answer:

- Doppler shift $f_d = 2vf_c/c$; at 2.462 GHz that is **16.4 Hz per m/s** of radial speed.
- We divided the `action` power spectrum by the `noaction` spectrum. **Above 18 Hz the ratio is 1.0** — the classes are indistinguishable there, that band is pure noise floor.
- **All class-discriminative information lies below 18 Hz.**
- Our worst-case Nyquist is **25.8 Hz**, median **34.1 Hz → 1.43×–1.90× headroom**.

> "So the 15× rate deficit versus UT-HAR costs us nothing measurable *on this task*, and I can show the spectrum. **But** — this only holds because walk and mixed are gross whole-body motions. A fast hand at 3 m/s is 49 Hz and would alias. For the 7-class collection with `drink` and `nod`, we must repeat this exact test *before* bulk recording. It takes ten minutes."

That answer converts a suspected fatal flaw into evidence of rigour **plus** a concrete plan.

Part 7 — Weaknesses: prepared answers

Likely attack	Your answer
"98% is meaningless, task is trivial"	**Agree, and show §15 first.** One scalar gets the same. Explain what the number *does* validate, and that the structural justification is §3/§4/§13/§16.
"Only 2 subjects"	True and it is our biggest limitation. Each fold trains on one person. ARIL has 1 subject, UT-HAR 6, Widar 16. We need ≥5 subjects, ≥2 days, ≥2 geometries — it is item 7 on our fix list.
"Only 322 samples"	Correct — ARIL 1,394, UT-HAR 4,477, Widar 258,000. Plus we lost 13% to the `--rate 100` default, recoverable immediately. `--step 1.0` overlap would roughly double the rest.
"Why not use phase?"	Part 1 — CFO/SFO/timing, unsynchronised USRP. Widar's phase trick needs ≥2 antennas; we are SISO. ARIL discards phase even *with* GPS sync.
"Why not just use ResNet, it's proven?"	19.6× the parameters and 52 forward passes per window for equal-or-worse accuracy on our data. Also our dataset is 322 windows — an 11 M-parameter model on that is asking to overfit.

Likely attack	Your answer
"Your attention is decoration"	§13: peak/uniform 2.22×, correlation +0.346 with motion energy over 182 windows. Measured, not assumed.
"Bidirectional GRU can't work in real time"	Correct. We classify complete recorded windows offline, so it is valid here. For streaming we would switch to unidirectional or a sliding buffer — a known, accepted trade-off.
"Your two notebooks aren't comparable"	**Agree — say it before she does.** Different windowing (497 vs 322 windows, ~1.5 s vs exactly 2 s). What *is* comparable: parameters, forward passes, protocol cleanliness.
"Training accuracy hits 1.000 in 5 epochs — overfitting"	Yes, capacity far exceeds this task. Val stays 0.905–1.000 and test 98.4%, so it is not hurting *here*, but it is more evidence the task is too easy — same conclusion as §15.
"Why 2 s windows?"	Long enough to contain a full squat/bend cycle (~2–3 s per the capture protocol), short enough to give 322 samples from 752 s of capture. It is also what ImgFi and most CSI-HAR work uses, so it is comparable.
"Why did you drop rows?"	Exactly one truncated final row per file — the flowgraph was Ctrl+C'd mid-write. 0.015% of cells. We drop it explicitly rather than let NaN propagate.

Part 8 — Rapid-fire Q&A bank

On CSI / the data

- *How many subcarriers and why?* 52 — 802.11a/g 20 MHz, 64 FFT slots minus 12 null (DC+guards).
- *Sampling rate?* 51.5–77.6 Hz, mean 66.4. Passive sniffing, so it is whatever the link emits and `gr-ieee802-11` can decode (legacy OFDM only; 802.11n HT frames are undecodable).
- *Frequency?* 2.462 GHz, channel 11, 20 MHz bandwidth.
- *How many antennas?* One. SISO. That is why no phase-difference features.
- *Sample dimensions?* 52 × 200 = 2 s at the 100 Hz interpolation grid.
- *How many samples?* 322 windows: 168 action / 154 noaction.
- *Why interpolate onto a uniform grid?* Packets arrive irregularly (median gap 12 ms, worst 464 ms). Slicing by packet count would give windows of inconsistent physical duration — which is exactly what the ImgFi notebook does.

On the model

- *Total parameters?* 569,603. GRU is 87% of that.
- *Input to Conv1d?* 52 channels (subcarriers), sliding over time.
- *Kernel size 5 = how long?* 50 ms at the 100 Hz grid.
- *GRU sequence length?* 50 steps, each 40 ms → 2 s total.
- *GRU output width?* 256 = 128 hidden × 2 directions.
- *Attention parameters?* 257 (a single Linear(256→1)).
- *Inference time?* 0.55 ms per window, one forward pass.
- *Why GRU over LSTM?* ~25% fewer parameters at 50 steps; dataset is tiny.
- *Why dropout 0.5?* 322 windows — aggressive regularisation is deliberate.

On evaluation

- *Split?* Leave-one-subject-out, both directions, report the worse.
- *Why not random?* Adjacent windows from one take are near-duplicates → leakage → fake 99%.
- *Validation?* 15% of the *training* subject's windows, split by window. Test never used for selection.
- *Result?* 98.352% (amrith) / 98.571% (rishiikesh) → headline **98.35%**.
- *Baseline?* One scalar + one threshold: 98.35% / 99.29%. **The deep model does not beat it.**

On ImgFi

- *What is RT/RP?* $RT[i,j]=|x_i-x_j|$ — each pixel compares the signal at two moments, so the basic feature unit is a time-to-time relationship.
- *Why did the paper pick RT?* Best of its 5 methods (~99.8%) and cheapest to compute.
- *Why didn't you use it?* §3 — 52 subcarriers carry ~4 independent dimensions, so the ×52 image expansion is mostly duplication; and it costs 52 forward passes per window.
- *Did your ImgFi notebook follow the paper?* No — it uses ImageNet-pretrained ResNet-18, whereas the paper's central claim is a lightweight 3-conv CNN. Be upfront.

Part 9 — What to show, in order

Walk the notebook top to bottom; it is deliberately ordered as an argument.

Show	Section	The one sentence to say
1	§1 raw CSI + rate table	"This is the raw sensor output; we measure 66 Hz, and duration comes from timestamps, not row counts."
2	§1 heatmap	"Vertical stripes = one event hitting all 52 subcarriers together."
3	§2 window + RPs	"This is the model's input, and this is what ImgFi would turn it into."
4	§2 six-subcarrier RP strip	"These six images are near-duplicates — that motivates the next section."
5	**§3 effective rank**	"≈4 independent dimensions, not 52. This is why we process jointly."
6	**§4 spectral ratio**	"All class information is below 18 Hz; our Nyquist is 25.8 Hz. The rate is sufficient — measured, not assumed."
7	§5 dataset + the 13% disclosure	"322 windows, balanced, LOSO. And here is a mistake we found ourselves."
8	§7 shape trace + parameters	"52 subcarriers are the channel dimension — here is the tensor at every stage."
9	§10–12 results + confusion	"98.35% headline, worse of two folds."
10	**§13 attention**	"2.22× non-uniform, +0.346 correlation with motion energy — it is doing something."
11	§14 per-take	"`stand` is hardest, not `mixed` — here is why we think that is."
12	**§15 baseline**	"And here is the honest part: one scalar does just as well."
13	§16–17 efficiency + comparison	"19.6× fewer parameters, 1 pass instead of 52 — and here is why the accuracies aren't directly comparable."
14	§18 limitations	"These are the six things wrong with this dataset."

Part 10 — The closing statement

> "The pilot did exactly what a pilot is for. The capture chain works, the evaluation is leakage-free and stricter than ARIL's or UT-HAR's, and we found and fixed real problems — the 13% window loss, the normalisation leak, the test-set model selection.
>
> What we deliberately do **not** claim is that 98.4% proves our architecture. A single threshold matches it, because separating moving from still is a one-scalar problem. What we do claim is structural: cross-subcarrier information exists and we exploit it, our sampling rate provably covers the informative band, attention is measurably active, and we do it with 0.57 M parameters and one forward pass instead of 11.2 M and 52.
>
> The next step is the 7-class collection, where a single energy scalar cannot work — and before we record it, we will run the ten-minute spectral test on `nod` and `drink`, because that is the one place this hardware is most likely to fail."

Appendix — every number, one place

Quantity	Value
Subcarriers	52 (802.11a/g, 20 MHz, 64 FFT − 12 null)
Carrier / channel	2.462 GHz, ch 11
Antennas	1 (SISO)
Packet rate	51.5–77.6 Hz, mean 66.4
Median inter-packet gap / worst	12 ms / 464 ms
Window	2 s → 52 × 200 (100 Hz grid)
Windows	322 (168 action / 154 noaction), 52.2% action
Per subject	amrith 182, rishiikesh 140
Per take	walk 87, mixed 81, stand 81, sit 73
Windows lost to `--rate 100`	48 of 370 (13.0%)
Truncated rows	1 per file (0.015% of cells)
Mean |subcarrier corr| (action)	0.38–0.45
Effective rank	**3.8 (action) / 11.9 (noaction)** of 52
Discriminative band	**< 18 Hz**
Nyquist (worst / median)	25.8 / 34.1 Hz → 1.43× / 1.90× headroom
Doppler scaling	16.4 Hz per m/s
Parameters	569,603 (CNN 58,176 · GRU 494,592 · attn 257 · head 16,578)
ResNet-18 comparison	~11.18 M → 19.6× larger
Inference	0.55 ms/window, 1 pass (vs 52)
LOSO accuracy	**98.352% (amrith) / 98.571% (rishiikesh)**
F1 (action)	0.984 / 0.987
Per-take accuracy	walk 1.000, sit 1.000, mixed 0.988, **stand 0.951**

Quantity	Value
Attention peak/uniform	2.22×
Attention–energy correlation	+0.346
1-feature baseline	**98.35% / 99.29%**
Motion energy	action 0.01452, noaction 0.00346 → **4.20×**
ImgFi notebook windows	497 (270 action / 227 noaction)
ImgFi notebook accuracy	95.94% final, 97.79% best (amrith only)